



Guseyn Ismayylov

SENIOR FULL-STACK SOFTWARE ENGINEER · TECHNICAL LEADER

LEGAL NAME: GUSEIN ISMAIYLOV · REMOTE · GEORGIA 🇧🇪

guseyn@guseyn.com [+995 511 458 167](tel:+995511458167) [Telegram @guseyn](https://t.me/guseyn) guseyn.com

[GitHub](#) · [StackOverflow](#) · [LinkedIn](#) · [Instagram](#)

Founder, framework author, and Senior Full-Stack Engineer with **10 years** in production software. I build things I care about — tools for musicians, declarative web frameworks, and high-performance backends — and back that up with a decade of fintech, ed-tech, and enterprise work. Applied Mathematics background; writing on my blog guseyn.com. Occasionally use LLM tools as a research aid to explore ideas and sanity-check theories — not as a substitute for engineering.

Ventures & Passions

- [Unison](#) (*Founder / CTO, 2019 – Present*) — Platform that turns plain-text descriptions into sheet music. Long-running passion project at the intersection of music, notation, and software. Stack: Node.js, MongoDB, SVG, Markdown, EHTML. Working toward open-sourcing *Unilang*, a language for describing music in words.
 - [InstruxMusic](#) (*Tech Lead, 2024 – Present*) — Helping music schools and instructors run their business: scheduling, lesson tools, video chat, Stripe payments, brand builder. Built the core platform solo in under 3 months using my own frontend and backend tooling; involved in product, pricing, and business decisions.
 - [EHTML](#) (*Creator*) — Declarative frontend framework: custom HTML elements that handle routine client-side behaviour so you write less JavaScript. Used in Unison, InstruxMusic, and open source. [EHTML 3 Demo \(2026\)](#).
 - [Nodes.js](#) (*Author*) — Procedural Node.js backend framework with Cluster API over HTTP/2. The backend I reach for when I want speed and control without framework bloat.
 - **Music & writing** — Guitar and ukulele covers; working on an online book, *Guitar by Notes* ([sample chapter](#)).
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Skills

- **System Architecture:** Distributed Systems, Microservices, Event-Driven Architecture (EDA), Multi-tenant SaaS, High-Load MVPs, System Design, Scalability, High Availability & Reliability Engineering.
 - **Backend Ecosystem:** Node.js (Express, Fastify, NestJS), Nodex.js (Author), JavaScript/TypeScript, Java (Spring Boot), PHP, Python (FastAPI), NX Monorepos, RESTful APIs, GraphQL, Serverless Architecture.
 - **High-Performance UI/UX:** Web Components, EHTML (Founder), React, Next.js, React Server Components (RSC), React Context, React Query, Tailwind CSS, Material UI, Zustand, Angular, Real-time SVG/Canvas Rendering, Shadow DOM, Micro-frontends, HTML5/CSS3, Vite, Esbuild.
 - **Data Engineering & Storage:** PostgreSQL, MySQL, MariaDB, Redis (Caching, Pub/Sub), ClickHouse (OLAP), Snowflake, Supabase, Drizzle ORM, DynamoDB, Prisma, Apache Kafka, RabbitMQ, MongoDB, Elasticsearch, Database Optimization.
 - **Cloud Platforms & DevOps (IaC):** AWS (EC2, ECS, S3, RDS, Lambda, DynamoDB, CloudWatch), GCP (Cloud Run, App Engine, BigQuery, VPC), Azure, DigitalOcean, Docker, Kubernetes (K8s), Terraform, GitHub Actions, GitLab CI/CD, Infrastructure as Code (IaC).
 - **Observability & Telemetry:** OpenTelemetry, Prometheus, Grafana, New Relic, Graylog, Datadog, Distributed Tracing, Log Aggregation, Full-Stack Monitoring.
 - **Protocols & Integrations:** WebRTC (Xirsys), WebSockets, Stripe API (Payment Infrastructure), Postmark (Email Delivery), Azure AD (OAuth2/OIDC), Secure Identity Management.
 - **Testing & Quality Assurance:** Unit testing (JUnit 5, Jest, Vitest, pytest, PHPUnit), integration & API testing (Postman, Newman, Supertest, REST Assured), E2E testing (Playwright, Cypress), mocking (Mockito, Sinon, Jest mocks), TDD/BDD, contract testing, Testcontainers, coverage & reporting (Allure, Istanbul), CI test gates, load/performance testing.
 - **AI Integration & Tools:** LLM Middleware, AI Developer Tools (Claude Code, Codex, Cursor IDE, GitHub Copilot, Gemini), n8n.
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Work Experience

InstruxMusic — Senior Full-Stack Developer, Tech Lead

NOV 2024 – PRESENT · REMOTE · USA

- Architected and accelerated the launch of a high-load ed-tech ecosystem, delivering a production-ready MVP in under 3 months using a streamlined stack of Node.js, PostgreSQL, Docker, and DigitalOcean.
- Engineered a scalable multi-tenant scheduling engine and automated invoicing system, integrating Stripe Payments to establish seamless transactional workflows and eliminate manual billing overhead.
- Developed a proprietary, low-latency video conferencing solution from scratch using vanilla Web Components and Xirsys (STUN/TURN), bypassing heavy third-party framework dependencies to achieve highly efficient real-time streaming, chat, and data channel communication.
- Designed high-deliverability notification pipelines by implementing Postmark, significantly cutting down user communication friction through automated transactional workflows.
- Drive technical strategy and product vision in alignment with business scaling and pricing frameworks, managing execution speed while maintaining a high-performance, low-bloat architecture.

[View Recommendations](#)

Congrify — Full Stack Senior Software Engineer, Tech Lead

OCT 2023 – OCT 2024 · REMOTE · GERMANY/ITALY

- Directed technical strategy and system observability by architecting a specialized monitoring system using Grafana and AWS CloudWatch, enabling real-time detection of payment transaction anomalies and resolving critical infrastructure bottlenecks.
- Reduced AWS Lambda operational costs by 20% through targeted workflow optimizations, memory tuning, and improved data-processing logic across serverless environments handling Java, Python, and Node.js runtimes.
- Spearheaded the modernization of legacy microservices, dramatically improving concurrency patterns, query execution, and high-load stability by migrating complex datasets to ClickHouse (OLAP) and Snowflake.
- Enhanced large-scale data ingestion and asynchronous messaging pipelines by leveraging AWS SQS and robust Pub/Sub architectures, ensuring zero-loss system reliability and high availability under heavy transactional surges.
- Engineered a real-time streaming analytics layer alongside low-latency data tiers utilizing DynamoDB and S3 to power user-facing payment intelligence dashboards.
- Led an end-to-end frontend redesign and streamlined communication systems by integrating AWS SES for transactional mailings and optimizing the React ecosystem with advanced caching and sophisticated state management.

FxPro — Senior Software Engineer

NOV 2022 – OCT 2023 · LIMASSOL, CYPRUS

- Orchestrated infrastructure for 60+ microservices using Node.js (NestJS) and Java, delivering a 20% improvement in system reliability and a 15% reduction in operational downtime.
- Created a centralized Authentication Service with Azure Active Directory integration and successfully migrated payment configurations for 20+ global systems across 50+ countries.
- Optimized data consistency and performance across distributed services by leveraging Redis for caching and PostgreSQL for mission-critical financial data.
- Streamlined CI/CD pipelines using Artifactory in GitLab, reducing deployment cycles by 25% and ensuring high-availability across global environments.
- Implemented full-stack observability across 25+ services using New Relic, Graylog, and Grafana, enabling real-time bottleneck identification and rapid incident response.

[View Recommendations](#)

Cognitran — Senior Software Engineer

MAR 2021 – OCT 2022 · KRAKOW, POLAND

- Developed mission-critical, offline-first automotive diagnostic software and automated technical documentation pipelines for enterprise clients including Ford and Jaguar.

Huawei — Senior Software Engineer

JAN 2020 – MAR 2021 · MOSCOW, RUSSIA

- Designed high-performance cloud infrastructure components, optimized internal developer tooling, and crafted high-throughput custom data parsers.

US Mobile — Senior Software Engineer

FEB 2019 – JAN 2020 · REMOTE · USA

- Helped design technical solutions and created a roadmap for migrating infrastructure to Google Cloud Platform (GCP).
- Successfully oversaw the implementation of this replatforming, ensuring a smooth and efficient transition.
- Spearheaded the full automation of report processing from third-party sources, utilizing GCP services to significantly improve data processing and operational efficiency.

Grid Dynamics — Intern to Senior Software Engineer

DEC 2015 – JAN 2019

- Progressed from Intern to Senior Engineer, delivering high-impact cloud-native R&D internal tooling for major enterprise clients including Google, Raymond James, and Home Depot.

Open Source

[EHTML](#), [Nodes.js](#), [e-ui](#), [Mathematics of Neural Networks](#)

Certifications

[Oracle Certified Associate Java SE 8 Programmer](#)

Education

Applied Mathematics and Computer Science — *Saratov State University, 2012 - 2016*

[Bachelor's Thesis: Cost Minimization of Energy on the Control of Satellite's Angular Motion \(in Russian\)](#)